

Power Series Methods for Differential Equations

Power Series: A Fundamental Tool for Solving Differential Equations

Historical Perspectives, Mathematical Foundations, and Modern Applications

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September 5, 2026



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Historical Development and Evolution

Taylor polynomial approximation: Let f be a function with derivatives of order n at the point x_0 . Then there exist one and only one polynomial P of degree $\leq n$.

$$\begin{aligned} P_n(x) &= f(x_0) + f'(x_0)(x - x_0) + \cdots + \frac{f^{(n)}(x_0)}{n!}(x - x_0)^n \\ &= \sum_{i=0}^n \frac{f^{(i)}(x_0)}{i!}(x - x_0)^i \end{aligned}$$

The values of P_n and f , together with their derivatives, agree at x_0 .

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Interpolation $\longrightarrow$ $\left\{ \begin{array}{l} \text{Lagrange: values of } f \\ \text{Hermite: values + derivatives} \\ \text{Taylor: derivatives at a single point} \end{array} \right.$



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A function f is said to be analytic on an interval $(x_0 - r, x_0 + r)$ if f has a power-series expansion in this interval,

$$f(x) = \sum_{n=0}^{\infty} a_n(x - x_0)^n,$$

convergent for $|x - x_0| < r$.

$$f(x) = \sum_{k=0}^n \frac{f^{(k)}(x_0)}{k!}(x - x_0)^k + E_n(x).$$



Theorem

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Theorem (Sufficient condition for the convergence of a Taylor series)

If f is infinitely differentiable on an open interval

$$I = (a - r, a + r),$$

and if there exists a positive constant A such that

$$|f^{(n)}(x)| \leq A^n, \quad n = 1, 2, 3, \dots$$

for every $x \in I$, then the Taylor series generated by f at a converges to $f(x)$ for every $x \in I$.

Why are power series important?

We solve first-order linear differential equations, linear differential equations with constant coefficients of any order.



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$$y'' = q(x)y.$$

For any function (say, a continuous one) one of the equations of physics for which, to date, only numerical or asymptotic solutions can be obtained, or for which sometimes not even those are available would be completely solved.

Conditions for Finding Solutions of Ordinary Differential Equations Using Power Series

If the coefficients of a homogeneous linear differential equation

$$y_n(x) + P_1(x)y_{n-1}(x) + \cdots + P_n(x)y(x) = 0$$

are analytic in an interval $(x_0 - r, x_0 + r)$, then it can be shown that there exist n independent solutions $u_1, \cdots, u_n$, each of which is analytic on the same interval.



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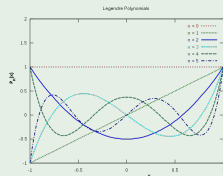
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Example (The Legendre equation)

$$(1 - x^2)y'' - 2xy' + (\alpha(\alpha + 1))y = 0$$

$$y'' - \frac{2xy'}{1 - x^2} + \frac{\alpha(\alpha + 1)y}{(1 - x^2)} = 0$$

$$|x| < 1$$



Historical Development - The method of Frobenius

If either $P_1, P_2, \dots, P_n$ is not analytic near x_0 , power-series solutions valid near x_0 may or may not exist.

Example

Find a power-series solution

$$x^2 y'' - y' - y = 0$$

near $x_0 = 0$. If we assume that a solution

$$y = \sum_{n=0}^{\infty} a_n x^n$$

exists and substitute this series in the differential equation we are led to the recursion formula

$$a_{n+1} = \frac{n-1}{n+1} a_n$$

The ratio test shows that this power series converges only for $x = 0$.



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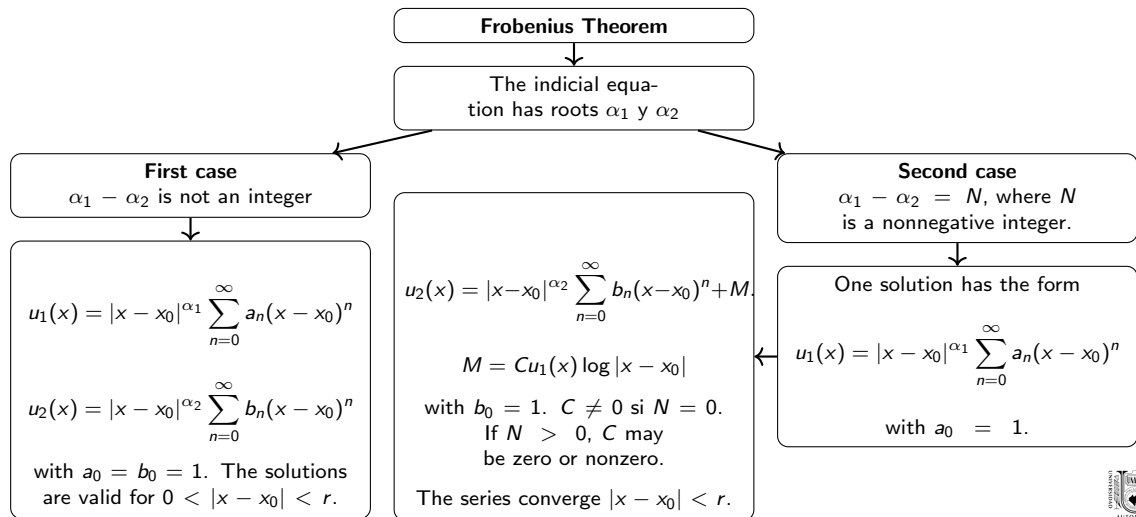
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In 1873 the German mathematician Georg Frobenius (1849–1917) developed a useful method for treating such equations. Frobenius' theorem splits into two parts, depending on the nature of the roots of the quadratic equation $t(t+1) + P(x_0)t + Q(x_0) = 0$. This quadratic equation is called the indicial equation.

The method of Frobenius



The Sturm-Liouville equation

- Joseph Fourier, who developed it in his work on the heat equation in the early 19th century. $\frac{\partial u}{\partial t} = \alpha^2 \frac{\partial^2 u}{\partial x^2}$.

$$u(x, t) = X(x)T(t)$$

$$X(x)T'(t) = \alpha^2 X''(x)T(t)$$

$$T'(t) = \alpha^2 \lambda T(t) \quad X''(x) = \lambda X(x)$$



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$$(p(x)y')' + q(x)y + \lambda r(x)y = 0, a < x < b,$$

The Sturm-Liouville problem

Consists not in solving the equation, but in finding values of the parameter λ for which the equation subject to a pair of boundary conditions,

$$\alpha y(a) + \beta y(a) = 0,$$

$$\gamma y(b) + \delta y(b) = 0$$

has a non-trivial solution, which must also be found.

Theorem

$$(p(x)u')' + q(x)u = \lambda r(x)u$$

Assume that on a finite interval $[a, b]$,

Equation

$$(pu')' + qu = 0$$

possesses a particular solution u_0 such that the functions $u_0^2 r$ and $1/(u_0^2 p)$ are continuous on $[a, b]$.

$$u = c_1 u_1 + c_2 u_2$$

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where c_1 and c_2 are arbitrary complex
constants,

$$u_1 = u_0 \sum_{k=0}^{\infty} \lambda^k \tilde{X}^{(2k)}$$

$$u_2 = u_0 \sum_{k=0}^{\infty} \lambda^k X^{(2k+1)}$$

with $\tilde{X}^{(n)}$ and $X^{(n)}$ being defined by the
recursive relations

Spectral Parameter Power Series (SPPS)

$$\tilde{X}^{(0)} \equiv 1, \quad X^{(0)} \equiv 1 \quad (1)$$

$$\tilde{X}^{(n)}(x) = \begin{cases} \int_{x_0}^x \tilde{X}^{(n-1)}(s) u_0^2(s) r(s) ds, & n \text{ odd,} \\ \int_{x_0}^x \frac{\tilde{X}^{(n-1)}(s)}{u_0^2(s) p(s)} ds, & n \text{ even,} \end{cases} \quad (2)$$

$$X^{(n)}(x) = \begin{cases} \int_{x_0}^x \frac{X^{(n-1)}(s)}{u_0^2(s) p(s)} ds, & n \text{ odd,} \\ \int_{x_0}^x X^{(n-1)}(s) u_0^2(s) r(s) ds, & n \text{ even.} \end{cases} \quad (3)$$

where x_0 is an arbitrary point in $[a, b]$ such that p is continuous at x_0 and $p(x_0) \neq 0$. Furthermore, both series converge uniformly on $[a, b]$.



$$L = \frac{d}{dx} p \frac{d}{dx} + q$$

Note that if $L u_0 = 0$, then L can be written in the factored form ¹

$$L = \frac{1}{u_0} \frac{d}{dx} p u_0^2 \frac{d}{dx} \frac{1}{u_0}.$$

¹Vladislav V. Kravchenko and R. Michael Porter. *Spectral parameter power series for Sturm–Liouville problems*, 2009.

Transmutation Operator

$$y'' + \lambda y = 0 \quad y'(0) = 0 \quad y'(\pi) + y(\pi) = 0$$
$$\cot(\sqrt{\lambda}\pi) = \sqrt{\lambda}$$

$$y'' = -\lambda y$$

$$y'' - q(x)y = -\lambda y$$

$$A = \frac{d^2}{dx^2} - q(x) \quad B = \frac{d^2}{dx^2}$$

$$AT = TB$$

$$ATu = TBu = -\lambda(Tu)$$

$$y = Tu$$

$$(p(x)y')' + q(x)y + \lambda r(x)y = 0$$

$$\alpha y(a) + \beta y'(a) = 0$$

$$\gamma y(b) + \sigma y'(b) = 0$$

$$T[u](x) = u(x) + \int_{-x}^x K(x, t)u(t) dt$$



Neumann Series

Given the equation

$$-y'' + q(x)y = \omega^2 y, \quad x \in [0, b].$$

two linearly independent solutions
have the form

$$c(\omega, x) = \cos(\omega x) + 2 \sum_{n=0}^{\infty} (-1)^n \beta_{2n}(x) j_{2n}(\omega x)$$

$$s(\omega, x) = \sin(\omega x) + 2 \sum_{n=0}^{\infty} (-1)^n \beta_{2n+1}(x) j_{2n+1}(\omega x)$$

- ① A partial sum of the series approximates the solution uniformly with respect to ω .
- ② The kernel of a transmutation operator is independent of the spectral parameter.²

²V.V. Kravchenko, Luis J. Navarro, S. M. Torba. *Representation of solution to the one-dimensional Schrödinger equation in terms of Neumann series of Bessel functions*, 2017

Spectral Parameter Power Series (SPPS) and Higher-Order Differential Equations

Let $y(\omega, x)$ denote the solution of the following Cauchy problem

$$y^{(n)} + p_1(x)y^{n-1} + p_2(x)y^{(n-2)} + \dots + p_n(x)y = \omega^n y, \quad x \in [0, b]$$

$$y(\omega, 0) = 1, y'(\omega, 0) = \omega, \dots, y^{(n-1)}(\omega, 0) = \omega^{n-1}$$

where $p_2, \dots, p_n$ are

complex-valued continuous functions of real variable on $[0, b]$ where $0 < b < \infty$, and $\omega \in \mathbb{C}$.

Examples

The solution $y(\omega, x)$ admits the representation

$$y(\omega, x) = \sum_{m=0}^{\infty} \frac{\alpha_m(x)}{m!} \omega^m.$$

The series converges uniformly with respect to $x \in [0, b]$ and uniformly on any compact subset of the complex plane with respect to ω .

The solution $y(\omega, x)$ admits the following representation

$$y(\omega, x) = \frac{1}{2\pi i} \int_C \gamma(x, t) e^{\omega t} dt$$

where for any fixed $x \in [0, b]$ the function $\gamma(x, t)$ as a function of the variable t is analytic outside a regular n -sided polygon Π_x with center at the origin and one of whose vertices being the point $t = x$.

3

³A. F. Leontiev, *Generalizations of exponential series* (in Russian), Moscow: Nauka, Home edition of physical and mathematical literature, 1981

- A transmutation operator for an equation of order greater than 2 is a problem that has long been studied by several mathematicians, such as Hachatrian, Fage⁴, Leontiev, Saknovich⁵, and Malamud.
- Analyzing some of these works, such as in Hachatrian's work where the coefficients must be analytical.
- The representation known in Leontiev's book. is more general.
- The transmutation operator for order 2 is obtained from this representation.

The following representation can be used as well

$$y(\omega, x) = e^{\omega x} + \frac{1}{2\pi i} \int_C \tilde{\gamma}(x, t) e^{\omega t} dt$$

⁴M.K. Fage, *Constructing transformation operators and solving a single moment problem for ordinary linear differential equations of arbitrary order*, UMN, 1957, Volume 12, Issue 1 (73), 240–245 (in Russian)

⁵Sakhnovich, Lev Aronovich *Conditions necessary for the presence of a transformation operator for fourth order equations* Advances in mathematical sciences volume.16, 5(101), 199-204. 1961(in Russian)



Borel transform

Based on the well-known results of complex variable about entire functions ⁶ we obtain the following result Let

$$y(\omega, x) = \sum_{m=0}^{\infty} \frac{\alpha_m(x)}{m!} \omega^m$$

be the representation SPPS of the solution $y(\omega, x)$ then outside Π_x the function $\gamma(x, t)$ admits the following series representation

$$\gamma(x, t) = \sum_{m=0}^{\infty} \frac{\alpha_m(x)}{t^{m+1}}.$$

In what follows by C we denote the counterclockwise oriented boundary of the square centered at the origin and with one of the vertex being $t_1 = (x + \delta) + i(x + \delta)$ with $\delta > 0$.

⁶R. P. Boas, *Entire functions*, vol.5, Academic Press, 2011



Neumann series Representation

Theorem

The solution $y(\omega, x)$ admits the following representation

$$y(\omega, x) = 2 \sum_{m=0}^{\infty} \left(\beta_m^{(1)}(x; \delta) e^{-i\omega(x+\delta)} + (-1)^{m+1} \beta_m^{(3)}(x; \delta) e^{i\omega(x+\delta)} \right) i_m(\omega(x+\delta)) \\ + 2 \sum_{m=0}^{\infty} \left(i^{m+1} \beta_m^{(2)}(x; \delta) e^{\omega(x+\delta)} + (-i)^{m+1} \beta_m^{(4)}(x; \delta) e^{-\omega(x+\delta)} \right) j_m(\omega(x+\delta))$$

j_m and i_m are spherical Bessel functions^a. These series converge uniformly with respect to x on $[0, b]$ and converge uniformly with respect to ω on any compact subset of the complex plane of the variable ω .

^a $j_n(z) = \sqrt{\frac{\pi}{2z}} J_{n+\frac{1}{2}}(z)$, $i_n(z) = i^{(-n)} j_n(iz)$

Inverse Problems and Power Series

We will apply power series to an inverse problems using the Sturm-Liouville equation. This type of problem is the foundation of modern seismology and quantum physics (discovering the earth structure or an atom's potential by measuring their vibrations or energy levels).

$$-y''(x) + q(x)y(x) = \lambda y(x)$$

Forward problem: You know the potential $q(x)$ and calculate the vibration frequencies or eigenvalues (λ).

Inverse Problem: The potential $q(x)$ is hidden (for example the variable density inside a material bar). We want to discover the coefficient $q(x)$ by taking measurements at the boundary ($x=0$).

$$q(x) = q_0 + q_1x + q_2x^2 + \dots$$

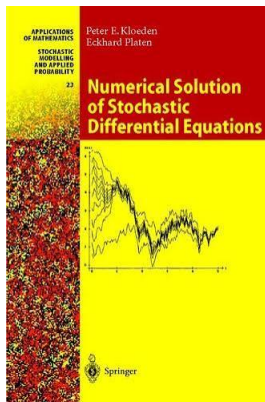


References and Future Directions

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References and Future Directions



Kloeden and Platen
*Numerical Solution of Stochastic Differential
Equations.*

Thank you for your attention!

